

# Database Documentation

The purpose of every table, and the complete column catalog.

This document answers two questions about the COMPASS PostgreSQL database.

**Part I** explains **why each table exists** — following one consultation transcript from arrival to the screens the doctor and the patient see. **Part II** is the **data dictionary**: every table, every column, its type, nullability, keys, and meaning.

|                 |                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| DATABASE        | prostatecancer_db_native · PostgreSQL 16 · schema public                                                     |
| SCOPE           | <b>15 application tables</b> in 5 groups, plus <code>alembic_version</code> (migration bookkeeping)          |
| SOURCE OF TRUTH | <code>app/Backend/models.py</code> — SQLAlchemy ORM definitions                                              |
| VERIFICATION    | Every column was diffed against the live database ( <code>information_schema.columns</code> ) — 0 mismatches |
| ASSEMBLED FROM  | <code>docs/architecture/DB_TABLES_ROLES.md</code><br><code>docs/architecture/DATABASE_SCHEMA.md</code>       |
| GENERATED       | 2026-07-09                                                                                                   |

Internal engineering documentation. Contains no protected health information.

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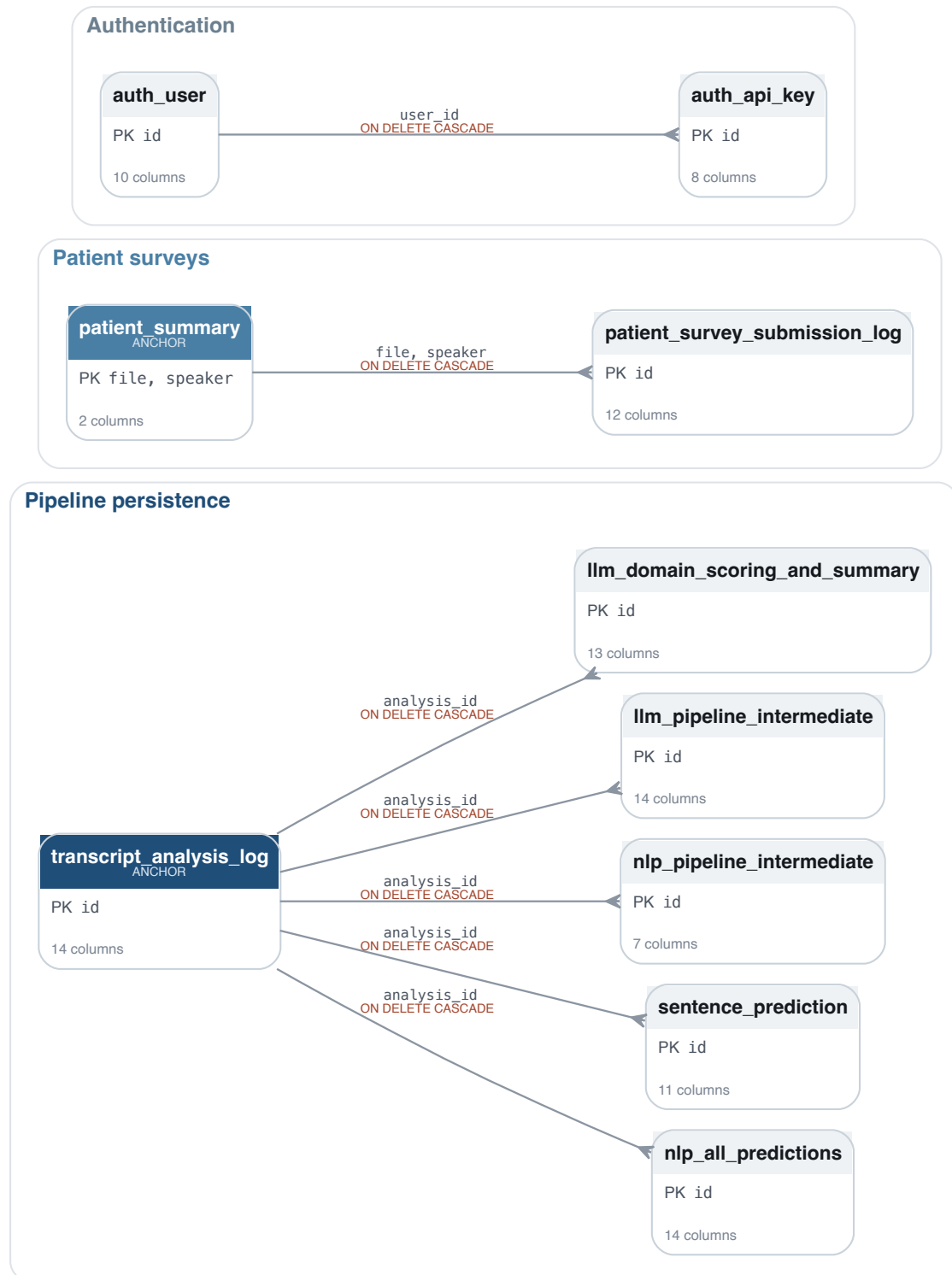
alembic\_version

Key indexes (beyond primary keys)

See Also

# Entity-relationship diagram

Generated directly from the live database catalog — tables, primary keys, foreign keys and their delete behaviour are read from `pg_constraint`, so the diagram cannot drift from the schema.



**Reading the diagram.** A crow's-foot marks the “many” end: one `transcript_analysis_log` row parents many rows in each pipeline table. The two **anchors** hold identity only; everything else points back at them. `ON DELETE CASCADE` means deleting the parent removes its children, so a pipeline run can be re-processed without orphans.

### Tables with no foreign keys

The remaining 6 tables stand alone — they are written by the behavior tracker, the rewrite log, the session recorder, or Alembic, and reference nothing.

|                                                                                            |                                                                                                     |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <div>SYSTEM</div> <div>alembic_version</div> <div>1 column</div>                           | <div>BEHAVIOR TRACKING</div> <div>doctor_behavior</div> <div>11 columns</div>                       |
| <div>OTHER</div> <div>doctor_rewrite_log</div> <div>9 columns</div>                        | <div>BEHAVIOR TRACKING</div> <div>patient_followup_survey_page_behavior</div> <div>12 columns</div> |
| <div>BEHAVIOR TRACKING</div> <div>patient_report_page_behavior</div> <div>11 columns</div> | <div>OTHER</div> <div>session_recording</div> <div>9 columns</div>                                  |

## Why each table exists

One consultation transcript, followed from the moment it arrives to the screens the doctor and the patient open. Each stage names the table it fills and what that table is for.

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## How one file lands in the DB — followed as a story

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One example file, `13511_13571_06302026.csv` (patient 13511 · doctor 13571), travels **transcript** → **NLP** → **AI** → **screen**, and at each step we follow **which drawer (table) gets what**. Real stored values are shown too. Example sentence: (**utterance 66, sentence 4**) = "now, it's a whole lot lesser..."

English mirror of `DB_TABLES_ROLES_KR.md`; if they diverge, the schema in `models.py` wins.

### First — "table = drawer"

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The DB is a **set of labeled drawers**, each holding a different kind of information. Processing one file spreads its data **across several drawers**.

- The drawers the pipeline creates are all tied together by one **analysis\_id (ticket number)**. (This file's ticket = 1.)

### Act 1 — the transcript arrives (de-identified)

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The raw `SID22_doc2_06302026.xlsx` (real consultation transcript) is not kept as-is: patient/doctor names are **hashed** into `13511_13571_06302026.csv`. → **Nothing in the DB yet**. Just file prep.

---

### Act 2 — create the analysis receipt → drawer ① `transcript_analysis_log`

---

When the pipeline starts on this file it makes **one "analysis receipt"** — the first drawer's first row. Its **id (=1)** becomes the **analysis\_id** — the ticket stamped on every drawer made afterward.

```
stored: id=1, patient_id=13511, doctor_id=13571, total_sentences=438, top_n=10, ai_overall_score=0.8
(filled later), processed=true (when done)
```

## Act 3 — NLP: split sentences and score them

### 3-1. split into sentences

Each utterance (one turn) is split into sentences, each getting an **address (utterance i, in-utterance i2)**. E.g. utterance 66 → 4 sentences, the 4th being **(66,4)** "now, it's a whole lot lesser..."

### 3-2. five graders score each sentence → drawer ② `nlp_all_predictions`

For every sentence, **5 models** (cp cancer-prognosis · le life-expectancy · ed erectile · inc incontinence · ius irritative-symptoms) score "probability this sentence is that topic". One row per sentence.

```
(66,4): pred_cp=0.781, pred_le=0.713, pred_ed=0.318, pred_inc=0.413, pred_ius=0.376 → both cp & le high
= the sentence spans two topics. (All 438 sentences go in this drawer.)
```

### 3-3. keep only each topic's top-10 + context → drawer ③ `sentence_prediction`

Each model keeps its **top-10 sentences**, with surrounding context; the focus sentence is marked `<main>...`  
`</main>`.

- (66,4) was picked into **both** cp and le top-10 → **2 rows** (one model=cp, one model=le).

```
(66,4) cp row: model=cp, pred_score=0.781, context="...85 to 90 percent...<main>now it's a whole lot
lesser...</main>..."
```

(Side drawer `nlp_pipeline_intermediate` is a "working-notes" drawer of step snapshots — for debugging.)

## Act 4 — AI (GPT-4o): summarize and score

### 4-1. score · extract · filter → drawer ④ `llm_pipeline_intermediate`

For each domain's top-10 candidates GPT-4o records: a 0–5 score + extracted estimate/treatment. One row per candidate. Which candidates were ultimately chosen is answered by `llm_domain_scoring_and_summary`, not by this table.

```
a cp candidate: domain=cp, ai_score=2, estimate=<missing>, score_explanation="...cancer
mortality/survival or metastasis risk..."
```

#### 4-2. pick a domain representative and rewrite for the patient → drawer ⑤

##### llm\_domain\_scoring\_and\_summary

Each domain picks **one representative** and rewrites its sentence into **patient-friendly text**. This drawer is the **final result shown on the patient/doctor screens**.

```
cp: ai_score=2, source_sentence="and they're essentially looking to ensure...",  
reformat_sentence="You and your doctor did not discuss this information." (patient-facing)
```

Finally the all-domain average (**0.8**) is written to receipt ①'s `ai_overall_score`, and `processed=true` is stamped.

#### Act 5 — make the patient-side parent key → drawer ⑥ `patient_summary`

A **parent key** (`patient_summary`: file, speaker) is created for patient-side data to reference. Later patient survey submissions reference it by FK (integrity · cascade · re-processing survival).

```
patient_summary = (13511_13571_06302026.csv, Patient_13511_13571_06302026)
```

#### Act 6 — the doctor opens the screen — which drawers feed it

The doctor dashboard **reads the drawers via the backend API**:

- **Sentence grid** = sentences from ③ `sentence_prediction`, scores from ⑤ `llm_domain...`, side by side.
- **Score summary / trajectory** = per-domain `ai_score` from ⑤.
- When the doctor **rewrites a sentence** → new drawer ⑧ `doctor_rewrite_log` gets one row (original, revised, time) per rewrite.

#### Act 7 — the patient opens the screen — which drawers feed it

- Report **summary sentence** = ⑤ `llm_domain...`'s `reformat_sentence` (patient-facing rewrite).
- First-visit **Risk Perception answers** (sliders/choices) → drawer ⑦ `patient_survey_submission_log` (`survey_type='risk_perception_2'`, + REDCap `post_risk_perception_2`).
- **Follow-up surveys** (SDM/DCS/satisfaction) → the same drawer ⑦ `patient_survey_submission_log` (+ REDCap if enabled).

#### Act 8 — drawers that quietly fill in the background (behavior · recording)

Every **click/navigation** logs one behavior row:



- Patient first-visit **report page** behavior → `patient_report_page_behavior` (report-only)
- Patient **survey page** behavior (follow-up SDM/DCS/satisfaction + first-visit Risk survey) → `patient_followup_survey_page_behavior`
- Doctor dashboard behavior → `doctor_behavior`
- Screen recording (replay) → `session_recording` → These are **research usability logs** (behavior/timing only)
  - separate from answer values (values live in ⑦ `patient_survey_submission_log`).

## Act 9 — the gatekeeper drawers (auth · access)

Drawers that guard who may see what:

- `auth_user` (accounts) · `auth_api_key` (API keys).
- (Plus `alembic_version` = a system drawer tracking the DB schema version.)

## Appendix — step-by-step code map (file : function)

| STEP                           | FILE                                                   | FUNCTION                                                                     |
|--------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------|
| NLP A1 preprocess              | <code>sentence_classification/preprocessing.py</code>  | <code>identify_doctor_speaker · filter_doctor_rows</code>                    |
| NLP A2 segment                 | <code>sentence_classification/segmentation.py</code>   | <code>segment_sentences</code>                                               |
| NLP A3 5-model<br>classify     | <code>sentence_classification/classification.py</code> | <code>classify_all_models</code>                                             |
| NLP A4 top-N select            | <code>sentence_classification/selection.py</code>      | <code>select_top_k_sentences</code>                                          |
| NLP A5 attach<br>context       | <code>sentence_classification/context.py</code>        | <code>add_context_column</code>                                              |
| AI B1 score                    | <code>ai_pipeline/scoring.py</code>                    | <code>run_scoring</code>                                                     |
| AI B2 extract                  | <code>ai_pipeline/extraction.py</code>                 | <code>run_extraction</code>                                                  |
| AI B3 filter                   | <code>ai_pipeline/filtering.py</code>                  | <code>filter_candidates</code>                                               |
| AI B4 select<br>representative | <code>ai_pipeline/selection.py</code>                  | <code>run_selection</code>                                                   |
| AI B5 reformat                 | <code>ai_pipeline/reformat.py</code>                   | <code>run_reformat</code>                                                    |
| AI orchestration               | <code>ai_pipeline/pipeline.py</code>                   | <code>run_domain_pipeline / run_ai_pipeline</code>                           |
| DB persist                     | <code>db/persistence_helper.py</code>                  | <code>persist_pipeline_results → _save_nlp_results / _save_ai_results</code> |

| STEP                    | FILE                       | FUNCTION |
|-------------------------|----------------------------|----------|
| DB persist (NLP tables) | app/Backend/persistence.py | save_all |

## Epilogue — the drawer map on one page

```
[receipt] transcript_analysis_log (ticket analysis_id=1)
| this ticket ties the pipeline drawers below together
├ ② nlp_all_predictions          5-model probs per sentence (all)
├ ③ sentence_prediction          per-topic top-10 sentences + context (shared sentence = several
rows)
├ nlp_pipeline_intermediate      NLP working notes
├ ④ llm_pipeline_intermediate    AI candidate score·extract·filter
└ ⑤ llm_domain_scoring_and_summary AI representative + patient rewrite ← what the screen shows

[patient key] patient_summary (file, speaker) ← parent anchor for patient survey submissions

[what the screens use]
doctor: reads ③+⑤, on rewrite → ⑧ doctor_rewrite_log
patient: reads ⑤'s reformat, on survey input → ⑦ patient_survey_submission_log
(first-visit Risk = risk_perception_2, follow-up = sdm/dcs/satisfaction)

[in the background] behavior:
patient_report_page_behavior·patient_followup_survey_page_behavior·doctor_behavior / recording:
session_recording

[gatekeepers] auth_user·auth_api_key
```

**One-sentence summary:** a file comes in → the **receipt** (①) makes a ticket → **NLP fills sentences/probs** (②③), **AI fills scores/patient summaries** (④⑤) → the **screen reads** ③⑤ for doctor/patient, while **what people entered/edited** (survey ⑦ [patient\\_survey\\_submission\\_log](#) · doctor rewrites ⑧) and **behavior logs** pile up separately.

# Table catalog

The data dictionary. Every table, every column: type, nullability, keys, and what the value means. Grouped by the role the table plays in the system.

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## Database Schema — Data Dictionary

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Complete reference for the COMPASS PostgreSQL database: **15 application tables** (plus `alembic_version`) in the `public` schema. Every table and every column is documented below. Canonical definitions live in `app/Backend/models.py`.

Native deployment: port `:5433`, database `prostatecancer_db_native`. Docker mode: `:5432`.

**How to read the column tables:** `Key` column — **PK** primary key, **FK**→`t` foreign key to table `t`, **U** part of a unique constraint. `Null` — whether the column may be empty.

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### The parent–child (anchor) pattern

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Two tables hold **identity only** and act as anchors that other tables reference by foreign key:

- `transcript_analysis_log` (PK `id`) — one row per pipeline run; every NLP/AI result row points back to it via `analysis_id`. Deleting a run cascade-deletes all its results.
  - `patient_summary` (PK `file`, `speaker`) — one row per patient; the patient's survey submissions reference it. This gives referential integrity, cascade delete, and survival across re-processing.
-

## Group 1 — Pipeline persistence (7 tables)

The NLP + AI pipeline output. All are tied to one run via `analysis_id` → `transcript_analysis_log.id`.

### `transcript_analysis_log` — pipeline run header · anchor

One row per analysis run of one transcript file. Parent of every pipeline result table.

| COLUMN                           | TYPE         | NULL | KEY       | DESCRIPTION                                                                      |
|----------------------------------|--------------|------|-----------|----------------------------------------------------------------------------------|
| <code>id</code>                  | integer      | no   | <b>PK</b> | Auto-increment run id. Used as <code>analysis_id</code> by all child tables.     |
| <code>patient_id</code>          | varchar(255) | no   |           | Patient identifier (hashed) parsed from the file name.                           |
| <code>total_sentences</code>     | integer      | no   |           | Number of sentences the transcript segmented into.                               |
| <code>top_n</code>               | integer      | no   |           | How many top sentences per model were selected (e.g. 10).                        |
| <code>context_window</code>      | integer      | no   |           | How many neighboring sentences of context were attached (default 3).             |
| <code>model_results</code>       | jsonb        | yes  |           | Per-model NLP result payload (validated JSON) kept for reference.                |
| <code>xlsx_data</code>           | bytea        | yes  |           | The exported results xlsx stored inline, so downloads work without a filesystem. |
| <code>source_filename</code>     | varchar(500) | yes  |           | Original uploaded file name.                                                     |
| <code>pipeline_started_at</code> | timestampz   | yes  |           | When the pipeline orchestrator began this file.                                  |
| <code>analyzed_at</code>         | timestampz   | yes  |           | When NLP results were saved (default now()).                                     |
| <code>ai_overall_score</code>    | double       | yes  |           | Mean AI score across domains, written when the AI stage finishes.                |
| <code>processed</code>           | boolean      | yes  |           | True once the full AI pipeline completed for this run.                           |
| <code>processed_at</code>        | timestampz   | yes  |           | When the AI stage completed.                                                     |
| <code>doctor_id</code>           | varchar(255) | yes  |           | Doctor identifier (hashed) parsed from the file name.                            |

### `nlp_all_predictions` — every sentence × 5 model scores

One row per sentence, carrying all five model probabilities. The full raw NLP output.

| COLUMN                   | TYPE    | NULL | KEY                                              | DESCRIPTION                       |
|--------------------------|---------|------|--------------------------------------------------|-----------------------------------|
| <code>id</code>          | integer | no   | <b>PK</b>                                        | Auto-increment row id.            |
| <code>analysis_id</code> | integer | no   | <b>FK</b> → <code>transcript_analysis_log</code> | The run this sentence belongs to. |

| COLUMN                       | TYPE         | NULL | KEY | DESCRIPTION                                           |
|------------------------------|--------------|------|-----|-------------------------------------------------------|
| <b>patient_id</b>            | varchar(255) | no   |     | Patient identifier (denormalized for fast filtering). |
| <b>sentence_index</b>        | integer      | no   |     | Global sentence position in the transcript.           |
| <b>utterance_index</b>       | integer      | no   |     | Utterance (turn) number the sentence came from.       |
| <b>sentence_in_utterance</b> | integer      | no   |     | Position of the sentence within its utterance.        |
| <b>speaker</b>               | varchar(255) | yes  |     | Speaker label (doctor/patient) of the sentence.       |
| <b>sentence_text</b>         | text         | yes  |     | The sentence text.                                    |
| <b>pred_cp</b>               | double       | yes  |     | Probability the sentence is about Cancer Prognosis.   |
| <b>pred_le</b>               | double       | yes  |     | Probability — Life Expectancy.                        |
| <b>pred_ed</b>               | double       | yes  |     | Probability — Erectile Dysfunction.                   |
| <b>pred_inc</b>              | double       | yes  |     | Probability — Urinary Incontinence.                   |
| <b>pred_ius</b>              | double       | yes  |     | Probability — Irritative Urinary Symptoms.            |
| <b>created_at</b>            | timestampz   | no   |     | Row insert time (default now()).                      |

(*pred\_\** are NLP probabilities in  $[0,1]$ , not scores.)

## sentence\_prediction — top-N selected sentences (per model)

The per-domain representative sentences shown to the doctor. One row per (sentence × model) that made a model's top-N.

| COLUMN                       | TYPE         | NULL | KEY                               | DESCRIPTION                                             |
|------------------------------|--------------|------|-----------------------------------|---------------------------------------------------------|
| <b>id</b>                    | integer      | no   | <b>PK</b>                         | Auto-increment row id.                                  |
| <b>analysis_id</b>           | integer      | no   | <b>FK→transcript_analysis_log</b> | The run.                                                |
| <b>patient_id</b>            | varchar(255) | no   |                                   | Patient identifier (denormalized).                      |
| <b>model</b>                 | varchar(10)  | no   |                                   | Domain/model this row is for: <i>cp/le/ed/inc/ius</i> . |
| <b>sentence_index</b>        | integer      | no   |                                   | Global sentence position.                               |
| <b>utterance_index</b>       | integer      | no   |                                   | Utterance number.                                       |
| <b>sentence_in_utterance</b> | integer      | no   |                                   | Position within the utterance.                          |

| COLUMN               | TYPE         | NULL | KEY | DESCRIPTION                                                                                      |
|----------------------|--------------|------|-----|--------------------------------------------------------------------------------------------------|
| <b>speaker</b>       | varchar(100) | yes  |     | Speaker label.                                                                                   |
| <b>sentence_text</b> | text         | yes  |     | The selected sentence text.                                                                      |
| <b>pred_score</b>    | double       | no   |     | This model's probability for the sentence (the ranking value).                                   |
| <b>context</b>       | text         | yes  |     | Neighboring sentences; the focus sentence wrapped in <code>&lt;main&gt;...&lt;/main&gt;</code> . |

## nlp\_pipeline\_intermediate — NLP step-state snapshots

One JSONB snapshot per NLP step for debugging / reproduction.

| COLUMN             | TYPE         | NULL | KEY                               | DESCRIPTION                                                                                |
|--------------------|--------------|------|-----------------------------------|--------------------------------------------------------------------------------------------|
| <b>id</b>          | integer      | no   | <b>PK</b>                         | Auto-increment row id.                                                                     |
| <b>analysis_id</b> | integer      | no   | <b>FK→transcript_analysis_log</b> | The run.                                                                                   |
| <b>patient_id</b>  | varchar(255) | no   |                                   | Patient identifier.                                                                        |
| <b>step</b>        | varchar(20)  | no   |                                   | NLP step name:<br><code>raw/filtered/sentences/top_by_model</code><br>(CHECK-constrained). |
| <b>payload</b>     | jsonb        | no   |                                   | The step's input/output data.                                                              |
| <b>row_count</b>   | integer      | yes  |                                   | Number of rows in the payload (quick size indicator).                                      |
| <b>created_at</b>  | timestampz   | no   |                                   | Insert time.                                                                               |

## llm\_pipeline\_intermediate — per-domain AI intermediate state

One row per AI candidate (domain × sentence) recording how the AI scored/filtered it.

| COLUMN                | TYPE         | NULL | KEY                               | DESCRIPTION                                                |
|-----------------------|--------------|------|-----------------------------------|------------------------------------------------------------|
| <b>id</b>             | integer      | no   | <b>PK</b>                         | Auto-increment row id.                                     |
| <b>analysis_id</b>    | integer      | no   | <b>FK→transcript_analysis_log</b> | The run.                                                   |
| <b>patient_id</b>     | varchar(255) | no   |                                   | Patient identifier.                                        |
| <b>domain</b>         | varchar(10)  | no   |                                   | Domain: <code>cp/le/ed/inc/ius</code> (CHECK).             |
| <b>step</b>           | varchar(20)  | no   |                                   | AI step; currently always <code>extraction</code> (CHECK). |
| <b>sentence_index</b> | integer      | no   |                                   | Global sentence position of the candidate.                 |

| COLUMN                   | TYPE       | NULL | KEY | DESCRIPTION                                                     |
|--------------------------|------------|------|-----|-----------------------------------------------------------------|
| <b>sentence_text</b>     | text       | yes  |     | Candidate sentence text.                                        |
| <b>context</b>           | text       | yes  |     | Surrounding context.                                            |
| <b>pred_score</b>        | double     | yes  |     | Upstream NLP probability of the candidate.                      |
| <b>ai_score</b>          | smallint   | yes  |     | GPT-4o score 0–5 for the candidate.                             |
| <b>score_explanation</b> | text       | yes  |     | The model's reasoning for the score.                            |
| <b>estimate</b>          | text       | yes  |     | Extracted numeric estimate (long text; TEXT to avoid overflow). |
| <b>treatment</b>         | text       | yes  |     | Extracted treatment context.                                    |
| <b>created_at</b>        | timestampz | no   |     | Insert time.                                                    |

## llm\_domain\_scoring\_and\_summary — final patient-visible AI output

One row per domain (a domain may have several rows when the treatment branches). This is what the patient/doctor screens display.

| COLUMN                    | TYPE         | NULL | KEY                               | DESCRIPTION                                                                            |
|---------------------------|--------------|------|-----------------------------------|----------------------------------------------------------------------------------------|
| <b>id</b>                 | integer      | no   | <b>PK</b>                         | Auto-increment row id.                                                                 |
| <b>analysis_id</b>        | integer      | no   | <b>FK→transcript_analysis_log</b> | The run.                                                                               |
| <b>patient_id</b>         | varchar(255) | no   |                                   | Patient identifier.                                                                    |
| <b>domain</b>             | varchar(10)  | no   |                                   | Domain: <a href="#">cp/le/ed/inc/ius</a> .                                             |
| <b>ai_score</b>           | integer      | yes  |                                   | Final 0–5 AI score for the domain.                                                     |
| <b>score_explanation</b>  | text         | yes  |                                   | Reasoning behind the score.                                                            |
| <b>extracted_estimate</b> | text         | yes  |                                   | The final extracted numeric estimate.                                                  |
| <b>treatment</b>          | text         | yes  |                                   | Treatment this row pertains to (for treatment-branched domains).                       |
| <b>source_sentence</b>    | text         | yes  |                                   | The consultation sentence the AI chose as the source.                                  |
| <b>source_context</b>     | text         | yes  |                                   | Context with <code>&lt;main&gt;</code> markers; the screen's "From your consultation". |
| <b>reformat_sentence</b>  | text         | yes  |                                   | The patient-friendly rewrite shown in the report.                                      |
| <b>source_filename</b>    | varchar(500) | yes  |                                   | Original file name (denormalized).                                                     |
| <b>created_at</b>         | timestampz   | yes  |                                   | Insert time.                                                                           |

## patient\_summary — patient parent key · anchor

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One row per patient. Holds only the identity; patient survey submissions FK to it.

| COLUMN         | TYPE         | NULL | KEY       | DESCRIPTION                                        |
|----------------|--------------|------|-----------|----------------------------------------------------|
| <b>file</b>    | varchar(255) | no   | <b>PK</b> | The patient's file name (patient+doctor+date key). |
| <b>speaker</b> | varchar(100) | no   | <b>PK</b> | The patient speaker label inside that file.        |

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## Group 2 — Behavior tracking (3 tables)

UI-interaction events (how the user navigated), not answer values. All three share `session_id`, `file`, `speaker`, `event_type`, `metadata`, `device_type`, `client_timestamp`, `created_at` plus per-area extras. `event_type` is CHECK-constrained to each area's vocabulary.

### patient\_report\_page\_behavior — first-visit report page behavior

Events on the first-visit **report** page (reading AI summaries). Report-only.

| COLUMN                        | TYPE         | NULL | KEY       | DESCRIPTION                                                                                                                                                                                               |
|-------------------------------|--------------|------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>id</code>               | integer      | no   | <b>PK</b> | Auto-increment row id.                                                                                                                                                                                    |
| <code>session_id</code>       | varchar(100) | no   |           | One id per page mount; groups a visit's events.                                                                                                                                                           |
| <code>file</code>             | varchar(255) | no   |           | Patient file.                                                                                                                                                                                             |
| <code>speaker</code>          | varchar(100) | no   |           | Patient speaker.                                                                                                                                                                                          |
| <code>event_type</code>       | varchar(30)  | no   |           | Interaction type:<br><code>page_view/topic_open/topic_close/evidence_open/evidence_close/summary_open/summary_close/rating_click/slider_moved/answer_changed/domain_submitted/session_end</code> (CHECK). |
| <code>domain</code>           | varchar(50)  | yes  |           | Domain the event is about ( <code>cp/le/ed/inc/ius</code> ); required for topic/evidence/rating events.                                                                                                   |
| <code>rating</code>           | smallint     | yes  |           | 1–5 rating (for <code>rating_click</code> ; CHECK 1–5).                                                                                                                                                   |
| <code>metadata</code>         | jsonb        | no   |           | Free-form event details (timing, screen, etc.); default {}.                                                                                                                                               |
| <code>device_type</code>      | varchar(20)  | yes  |           | <code>mobile/tablet/desktop</code> .                                                                                                                                                                      |
| <code>client_timestamp</code> | timestampz   | no   |           | When the event happened on the client.                                                                                                                                                                    |
| <code>created_at</code>       | timestampz   | no   |           | Server insert time.                                                                                                                                                                                       |

### patient\_followup\_survey\_page\_behavior — survey page behavior

Events on the survey pages: follow-up (SDM/DCS/Risk/Satisfaction) **and** the first-visit Risk survey.

| COLUMN                  | TYPE         | NULL | KEY       | DESCRIPTION            |
|-------------------------|--------------|------|-----------|------------------------|
| <code>id</code>         | integer      | no   | <b>PK</b> | Auto-increment row id. |
| <code>session_id</code> | varchar(100) | no   |           | Session grouping id.   |
| <code>file</code>       | varchar(255) | no   |           | Patient file.          |

| COLUMN                  | TYPE         | NULL | KEY | DESCRIPTION                                                                                                                            |
|-------------------------|--------------|------|-----|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>speaker</b>          | varchar(100) | no   |     | Patient speaker.                                                                                                                       |
| <b>event_type</b>       | varchar(30)  | no   |     | page_view/survey_step_view/survey_answer/survey_complete/session_end (+ the report-page vocabulary for the embedded Risk step); CHECK. |
| <b>survey_type</b>      | varchar(30)  | yes  |     | Which survey: sdm/dcs/risk_perception/satisfaction (CHECK). Required for survey_answer.                                                |
| <b>question_id</b>      | varchar(50)  | yes  |     | Question the event refers to. Required for survey_answer.                                                                              |
| <b>step_number</b>      | smallint     | yes  |     | Survey step index (only for survey_step_view).                                                                                         |
| <b>metadata</b>         | jsonb        | no   |     | Free-form details; default {}.                                                                                                         |
| <b>device_type</b>      | varchar(20)  | yes  |     | Device type.                                                                                                                           |
| <b>client_timestamp</b> | timestamptz  | no   |     | Client event time.                                                                                                                     |
| <b>created_at</b>       | timestamptz  | no   |     | Server insert time.                                                                                                                    |

## doctor\_behavior — doctor dashboard behavior

Events on the doctor consultation dashboard.

| COLUMN                  | TYPE         | NULL | KEY       | DESCRIPTION                                                                                                                                                                                   |
|-------------------------|--------------|------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>id</b>               | integer      | no   | <b>PK</b> | Auto-increment row id.                                                                                                                                                                        |
| <b>session_id</b>       | varchar(100) | no   |           | Session grouping id.                                                                                                                                                                          |
| <b>file</b>             | varchar(255) | yes  |           | Patient file (null on the dashboard list view).                                                                                                                                               |
| <b>speaker</b>          | varchar(100) | no   |           | Doctor identifier.                                                                                                                                                                            |
| <b>event_type</b>       | varchar(30)  | no   |           | page_view/view_change/patient_select/topic_select/sentence_select/rewrite_open/rewrite_input/rewrite_apply/rubric_open/rubric_close/rubric_score_lock/tour_open/tour_end/session_end (CHECK). |
| <b>target_type</b>      | varchar(20)  | yes  |           | What the event targets: patient/topic/sentence (CHECK).                                                                                                                                       |
| <b>target_id</b>        | varchar(255) | yes  |           | Identifier of the target (patient/topic/sentence).                                                                                                                                            |
| <b>metadata</b>         | jsonb        | no   |           | Free-form details; default {}.                                                                                                                                                                |
| <b>device_type</b>      | varchar(20)  | yes  |           | Device type.                                                                                                                                                                                  |
| <b>client_timestamp</b> | timestamptz  | no   |           | Client event time.                                                                                                                                                                            |
| <b>created_at</b>       | timestamptz  | no   |           | Server insert time.                                                                                                                                                                           |

## Group 3 — Authentication (2 tables)

### auth\_user — accounts

| COLUMN               | TYPE         | NULL | KEY       | DESCRIPTION                                                              |
|----------------------|--------------|------|-----------|--------------------------------------------------------------------------|
| <b>id</b>            | integer      | no   | <b>PK</b> | Auto-increment user id.                                                  |
| <b>username</b>      | varchar(150) | no   |           | Login name.                                                              |
| <b>email</b>         | varchar(255) | yes  | <b>U</b>  | Email (unique).                                                          |
| <b>password_hash</b> | varchar(255) | yes  |           | Hashed password (null for external auth providers).                      |
| <b>role</b>          | varchar(20)  | no   |           | <code>admin/user/readonly</code> (CHECK); default <code>user</code> .    |
| <b>is_superuser</b>  | boolean      | no   |           | Superuser flag; default false.                                           |
| <b>is_active</b>     | boolean      | no   |           | Whether the account is enabled; default true.                            |
| <b>auth_provider</b> | varchar(50)  | no   |           | <code>local</code> or an external provider; default <code>local</code> . |
| <b>created_at</b>    | timestampz   | yes  |           | Creation time.                                                           |
| <b>updated_at</b>    | timestampz   | yes  |           | Last update time.                                                        |

### auth\_api\_key — API keys

| COLUMN              | TYPE         | NULL | KEY                  | DESCRIPTION                                                                             |
|---------------------|--------------|------|----------------------|-----------------------------------------------------------------------------------------|
| <b>id</b>           | integer      | no   | <b>PK</b>            | Auto-increment key id.                                                                  |
| <b>user_id</b>      | integer      | no   | <b>FK</b> →auth_user | Owner (cascade delete).                                                                 |
| <b>key_hash</b>     | varchar(255) | no   |                      | Hashed API key (raw key never stored); compared with <code>hmac.compare_digest</code> . |
| <b>label</b>        | varchar(100) | yes  |                      | Human label for the key.                                                                |
| <b>is_active</b>    | boolean      | no   |                      | Whether the key is enabled; default true.                                               |
| <b>created_at</b>   | timestampz   | yes  |                      | Creation time.                                                                          |
| <b>expires_at</b>   | timestampz   | yes  |                      | Optional expiry.                                                                        |
| <b>last_used_at</b> | timestampz   | yes  |                      | Last time the key authenticated.                                                        |

## Group 4 — Other (3 tables)

### patient\_survey\_submission\_log — all patient survey answers

One row per submission. The single source of truth for **every** patient survey answer: follow-up surveys ([sdm/dcs/satisfaction](#)) and the first-visit Risk cognition survey ([risk\\_perception\\_2](#)). FK to [patient\\_summary](#).

| COLUMN                  | TYPE         | NULL | KEY                                         | DESCRIPTION                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|--------------|------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>id</b>               | integer      | no   | <b>PK</b>                                   | Auto-increment row id.                                                                                                                                                                                                                                                                                                                                    |
| <b>file</b>             | varchar(255) | no   | <b>FK</b> → <a href="#">patient_summary</a> | Patient file.                                                                                                                                                                                                                                                                                                                                             |
| <b>speaker</b>          | varchar(100) | no   | <b>FK</b> → <a href="#">patient_summary</a> | Patient speaker.                                                                                                                                                                                                                                                                                                                                          |
| <b>survey_type</b>      | varchar(50)  | no   |                                             | <a href="#">risk_perception_2</a> (first-visit Risk), <a href="#">sdm</a> , <a href="#">dcs</a> , <a href="#">satisfaction</a> , ...                                                                                                                                                                                                                      |
| <b>answers</b>          | jsonb        | no   |                                             | The submitted answers (shape depends on <a href="#">survey_type</a> ; <a href="#">risk_perception_2</a> nests domain→question_id).                                                                                                                                                                                                                        |
| <b>extra_data</b>       | jsonb        | yes  |                                             | Submission status { <a href="#">partial</a> : bool} — <a href="#">true</a> = auto-save progress, <a href="#">false</a> = completed submission (+ any client metadata).                                                                                                                                                                                    |
| <b>submitted_at</b>     | timestampz   | yes  |                                             | Submission time (default now()).                                                                                                                                                                                                                                                                                                                          |
| <b>redcap_synced</b>    | boolean      | yes  |                                             | Whether the row was pushed to REDCap; default false.                                                                                                                                                                                                                                                                                                      |
| <b>redcap_record_id</b> | varchar(255) | yes  |                                             | REDCap-side record id after sync.                                                                                                                                                                                                                                                                                                                         |
| <b>redcap_error</b>     | text         | yes  |                                             | Last REDCap sync error (for retry), if any.                                                                                                                                                                                                                                                                                                               |
| <b>sid</b>              | varchar(50)  | yes  |                                             | Un-hashed subject id (e.g. <a href="#">SID_22</a> ), recovered from the hashed speaker composite <a href="#">Patient_&lt;hashedPatient&gt;_&lt;hashedDoctor&gt;_&lt;date&gt;</a> via <a href="#">deid.py</a> ; also the REDCap <a href="#">record_id</a> . Indexed ( <a href="#">idx_survey_submission_sid</a> ). Nullable — the un-hash may not resolve. |
| <b>doctor</b>           | varchar(50)  | yes  |                                             | Un-hashed doctor number (e.g. <a href="#">doc2</a> ) from the same composite, carried as attribution state. Nullable for the same reason.                                                                                                                                                                                                                 |

## session\_recording — consultation audio/replay chunks

| COLUMN                | TYPE         | NULL | KEY       | DESCRIPTION                                                                                                                 |
|-----------------------|--------------|------|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>id</b>             | integer      | no   | <b>PK</b> | Auto-increment row id.                                                                                                      |
| <b>session_id</b>     | varchar(100) | no   |           | Recording session id.                                                                                                       |
| <b>chunk_index</b>    | integer      | no   |           | Ordering index of this chunk; default 0.                                                                                    |
| <b>file</b>           | varchar(255) | yes  |           | Patient file the recording belongs to.                                                                                      |
| <b>visit_type</b>     | varchar(20)  | yes  |           | Which visit produced it.                                                                                                    |
| <b>recording_data</b> | bytea        | yes  |           | The raw recorded bytes.                                                                                                     |
| <b>event_count</b>    | integer      | no   |           | Number of captured events in the chunk; default 0.                                                                          |
| <b>created_at</b>     | timestampz   | yes  |           | Insert time.                                                                                                                |
| <b>area</b>           | varchar(40)  | no   |           | Capture area:<br>patient_first/patient_first_report/patient_first_survey/patient_followup/doctor/physician/unknown (CHECK). |

## doctor\_rewrite\_log — AI-assisted sentence rewrites

One row per rewrite. Composite PK (**file**, **i**, **i2**, **time**) keeps every revision as a separate audit row.

| COLUMN                   | TYPE         | NULL | KEY       | DESCRIPTION                                                            |
|--------------------------|--------------|------|-----------|------------------------------------------------------------------------|
| <b>file</b>              | varchar(255) | no   | <b>PK</b> | Patient file.                                                          |
| <b>i</b>                 | integer      | no   | <b>PK</b> | Utterance index of the sentence.                                       |
| <b>i2</b>                | integer      | no   | <b>PK</b> | In-utterance index of the sentence.                                    |
| <b>time</b>              | timestampz   | no   | <b>PK</b> | Rewrite time — part of the PK so revisions don't overwrite each other. |
| <b>speaker</b>           | varchar(100) | yes  |           | Speaker label of the sentence.                                         |
| <b>original_sentence</b> | text         | yes  |           | The sentence before the rewrite.                                       |
| <b>revised_sentence</b>  | text         | yes  |           | The doctor's rewritten sentence.                                       |
| <b>score</b>             | double       | yes  |           | Score associated with the rewrite.                                     |
| <b>class</b>             | varchar(100) | yes  |           | Domain/class of the sentence.                                          |

## Group 5 — System (1 table)

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### alembic\_version

| COLUMN             | TYPE        | NULL | KEY       | DESCRIPTION                                                                     |
|--------------------|-------------|------|-----------|---------------------------------------------------------------------------------|
| <b>version_num</b> | varchar(32) | no   | <b>PK</b> | The migration revision the DB is currently at. Managed by Alembic, not the app. |

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### Key indexes (beyond primary keys)

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- Pipeline: `analysis_id` and `(patient_id, ...)` indexes on the child tables for fast per-run / per-patient reads.
- Behavior: `(session_id)`, `(file, event/survey)`, `(client_timestamp)` on each behavior table.
- Surveys: `(file, submitted_at)`, `(survey_type)`, `(speaker)` on `patient_survey_submission_log`.
- Auth: `(user_id)`, `(key_hash)` on api keys; `(username)` on users.

### See Also

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- [ARCHITECTURE.md](#) — system overview · [DB\\_TABLES\\_ROLES.md](#) — the story of one file's journey · [AI\\_PIPELINE\\_NLP\\_DB\\_TABLES.md](#) — pipeline→table mapping · [INDEX.md](#) — one-page overview.
- [app/Backend/models.py](#) — canonical SQLAlchemy definitions.